Has the number of smokers decreased in the last 20 years in England?

This research question aims at identifying a potential decreasing pattern for smoking in England from 1999 to 2019. This is key as smoking has been consistently associated with many different health conditions such as lung cancer and heart disease (Parmar et al., 2023). In spite of being an unhealthy habit that can be prevented, people still engage in smoking. Consequently, it is needed to understand what factors drive the start and maintenance of smoking to effectively intervene as early as possible.

One factor that has been claimed to influence smoking is glamourization in films (Barker et al., 2020). In England, 70% of top popular films from 1989 to 2008 had tobacco content. While stricter bans on smoking in public places came into effect in the early 2000s (BBC, 2015), there was still a 40% of tobacco content in films from 2009 to 2017 (Barker et al., 2020). This means that even though there is an improvement in one of the potential drivers for smoking, it is not known whether it is enough to reduce the behaviour.

From the 1990s different public health strategies have been implemented, such as warning imagery in the cigarette packages, or the introduction of nicotine replacement treatments, e.g., vaping. Smoking is now represented as not only something harmful for health but also unattractive and less glamorous than how had been portrayed in films. Assessing UK smoking trends from the 90s can give us an insight on whether public health strategies are actually working. This can be of interest to public health practitioners, policymakers, researchers, and the general population.

The Health Survey for England (HSE)* provides nationally representative estimates of smoking trends in the population. It has been carried out every year since 1990, and collects information on health status, including smoking behaviour. The data collection procedure consists of two stages: 1) the researcher visits the household to screen/interview the eligible person or fulfil an individual questionnaire, 2) nurse visits to take some measures such as blood samples. For the present study we will analyse the smoking pattern with the study number 4365 (1999 wave, n = 10259), and the study number 8860 (2019 wave, n = 10299).

*Further information available in the Health Survey for England web page.

Variables of interest

Datasets were reduced to 10 variables:

1999 (10259 x 10)
Pserial, booklet, cignow, cigdyal, kcigreg, kcigregg, weigh, year, id, smoke

2019 (10299 x 10)
SerialA, SCType, cignow_2019, cigdyal, KCigReg, kcigregg, weight, year, id, smoke

More information about item type and use in the current analyses can be found in the appendix.

The quantity of null values can be seen in Figure 1 and 2:

Figure 1. Nulls for 1999

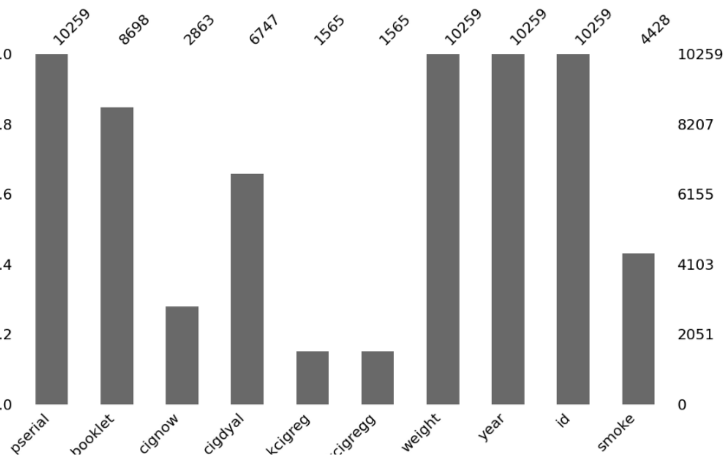
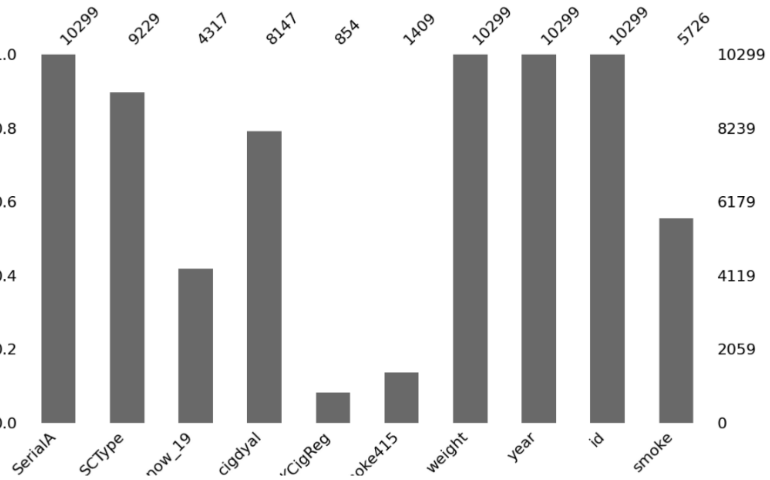


Figure 2. Nulls for 2019



Hypothesis:

H0: There are no significant differences in prevalence of smoking comparing smokers versus non-smokers in 1999 and 2019
H1: There are significant differences in prevalence of smoking comparing smokers versus non-smokers in 1999 and 2019

Analyses

The relation between smoking prevalence and year was analysed using a weighted logistic regression. Since we have a predictor variable that is categorical (time), and a categorical outcome (smoke), logistic regression analysis is a possibility to predict the probability of smoking. We can test if there are fewer people smoking in 2019 compared to 1999 taking into consideration weights.

In addition, we complemented this analysis by focusing on the smoking frequency in 1999 and 2019 within adults and young people separately (based on the booklet offered). The aim was to check whether among smokers there would be a decline tendency in the quantity of cigarettes smoked in 2019 compared to 1999. For adults, the predictor variable is time (categorical), but the outcome is continuous (number of cigarettes smoked a day). Since the distribution is very skewed, a non-parametrical test was used, Mann-Whitney U test. For young people, the predictor is time, and the outcome is ordinal, with 6 categories (frequency and amount smoked). Thus, a Mann-Whitney U test was also performed as it is adequate to handle ordinal variables.

Overall, the methodological choices to answer the research question were:

- Focus on smoking cigarettes instead of pipes or cigars, or a combination of the three. The reason is that only cigarettes was consistently asked across samples.
- In 2019 there was an item asking for self-report smoking status in children. However, kcigregg was preferred as it is the same item present in 1999.
- Chi square analysis is also an option to test a potential decrease pattern but logistic regression was chosen to better control for weights in python. In addition, the results are easier to interpret (odds).
- **Definition of "adult" vs "young people" depends on the instrument used.** The smoking module could be asked openly in an interview (in front of other household members) or in a self-completion format, with privacy. For 1999, participants that were +21-year-old were openly interviewed, young people (8 - 16) had to fulfil the booklet, and for +16- to 21-year-old participants the question format chosen depended on the research assistant. If the researcher judged that openly disclosing smoking information was sensitive or possibly leading to a biased answer,

then the participant was prompted to answer the smoking module with more privacy. However, in 2019, the smoking module was asked openly in an interview when the participant was +16, while if they were 8–15-year-old they were given the laptop to fulfil the questionnaire (CASI) with privacy. As the booklets are divided by age, “white adults” in booklet item (1999 dataset) and “Age +16” in SCType item (2019 dataset) were considered to define the adult category, while the rest of the classifications were part of the young category, with the “blue young adult” category (16 to 21) for 1999 dataset being excluded. The underlying reason is to ensure the smoking module is asked in the same way for the sample type analysed.

Results

Table 1. Prevalence estimates (weighted and unweighted)

Year	N[a]	Smokers	Weighted prevalence	Prevalence
1999	4428	1549	33.56	34.98
2019	5171	1262	26.2	24.41

Note. N[a]: valid sample: No missing values

Smoking prevalence was significantly lower in 2019 than in 1999 (β = -0.35, SE = 0.05, z = -7.86, p < .001). The odds of smoking were 30% lower in 2019 than in 1999 (OR = 0.70, 95% CI [0.64, 0.77]) (see Table 2).

For both groups, adults and youth, the median number of cigarettes smoked did not change. The Mann-Whitney U test indicated no statistically significant differences in smoking frequency between the two years (p>.05) (see Table 3).

Table 2. Main analysis: weighted logistic regression

	β	SE	Z (Wald)	p-value	95% CI
Const	-0.68	0.03	-21.55	<0.001	-0.75 to -0.62
Year2019	-0.35	0.05	-7.86	<0.001	-0.44 to -0.26

Table 3. Sensitivity analyses (frequency): smokers

	N (1999)	N (2019)	Median (1999)	Median (2019)	U	p-value	RBC
Adults[1]	1428	1197	10	10	880732, 5	.176	0,031
Youth[2]	172	34	2	2	2645	.297	-0.095

Notes. Adults[1]: cigarettes/day among smokers; Youth[2]: smoking frequency among smokers.

Discussion

Smoking became less prevalent in 2019 compared to 1999, as can be seen in figure 3. Indeed, the decline represents a substantial reduction in the proportion of people smoking over this time. This suggests that less people are exposed to the health risks mentioned above such as cancer, and it might indicate a potential overall effect of different public health strategies. However, the study design precludes us to infer any causality.

Interestingly, the quantity of cigarette smoking was similar in 1999 and 2019. This might indicate more resistance to reduce smoking, and underlines the importance of prevention campaigns to tackle this situation before it becomes an addiction.

Figure 3. Smoking prevalence

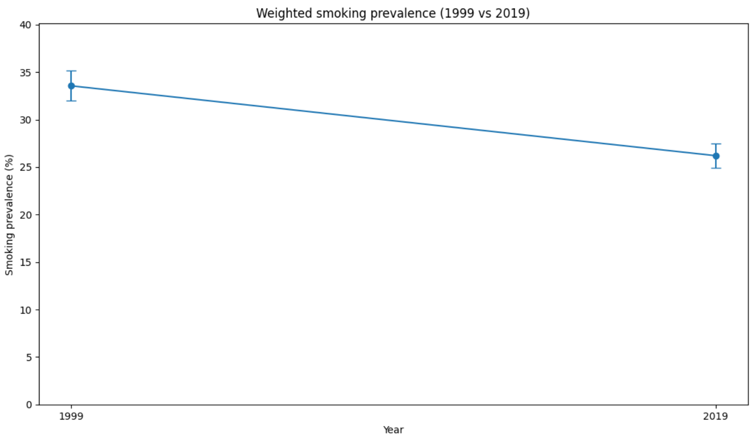


Figure 4. Quantity of cigarettes smoked per day by adults

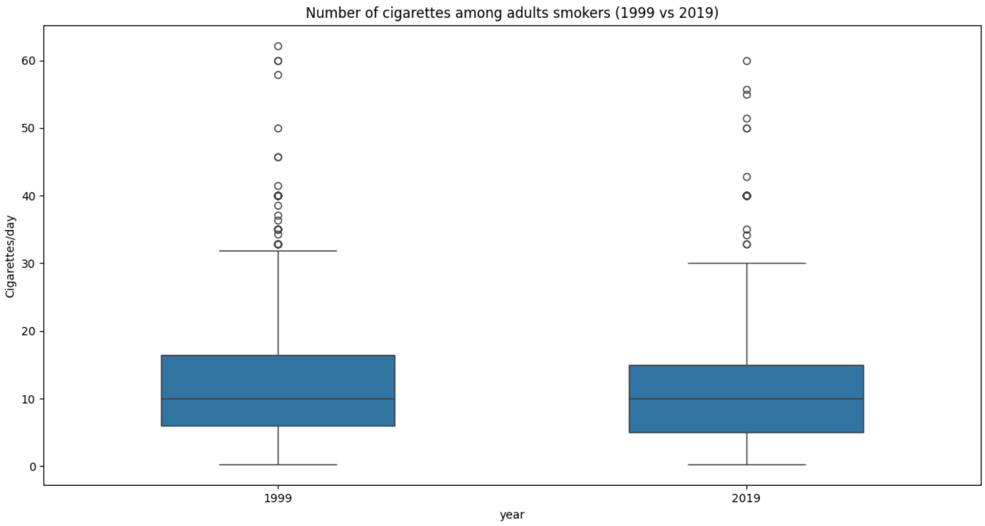
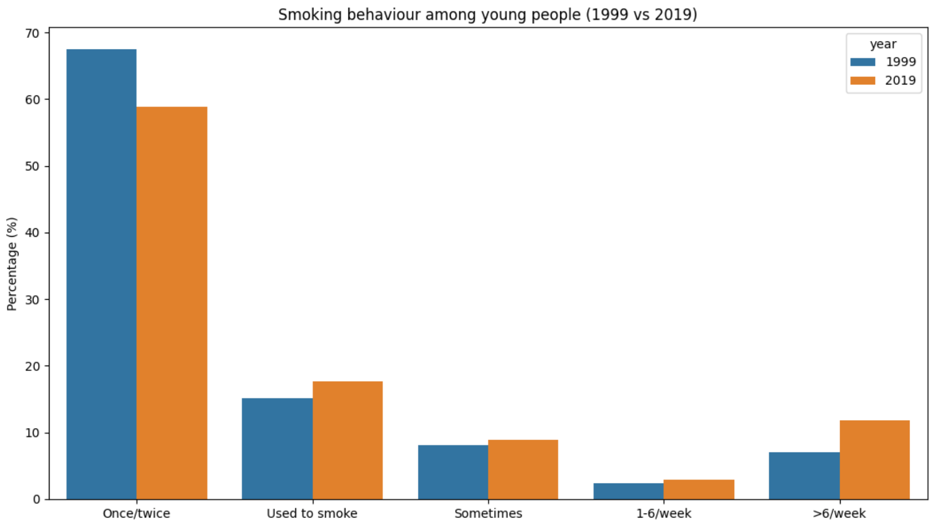


Figure 5. Percentage of smoking frequency in young people



References

Barker A, Cranwell J, Fitzpatrick I, et al. (2020). Tobacco and tobacco branding in films most popular in the UK from 2009 to 2017. *Thorax*. **75:**1103-1108.

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Parmar MP, Kaur M, Bhavanam S, Mulaka GSR, Ishfaq L, Vempati R, C MF, Kandepi HV, Er R, Sahu S, Davalgi S. (2023). A Systematic Review of the Effects of Smoking on the Cardiovascular System and General Health. *Cureus*. 15(4): e38073.

Appendix

The description of the chosen variables as they were in the **1999 dataset** are:

"**cignow**" (categorical): Whether smoke cigarettes nowadays.

Value = -9 Label = No answer/refused
Value = -8 Label = Don't know
Value = -6 Label = Schedule not obtained
Value = -2 Label = Schedule not applicable
Value = -1 Label = Item not applicable
Value = 1 Label = Yes
Value = 2 Label = No

Negative values were defined as Nas.

"**kcigregg**" (ordinal): Frequency of cigarette smoking (8-15s).

Value = -9 Label = No answer/refused
Value = -8 Label = Don't know
Value = -6 Label = Schedule not obtained
Value = -2 Label = Schedule not applicable
Value = -1 Label = Item not applicable
Value = 1 Label = Don't smoke cigarettes
Value = 2 Label = Smoke cigarettes, less than once a week
Value = 3 Label = Smoke cigarettes, once a week or more often

Negative values were defined as Nas. Values 2 and 3 were coded as smokers (1), while value 1 was coded for no-smokers (0).

"**smoke**" (categorical): Derived variable from cignow (adult) and kcigregg (children) for 1999 dataset. Whether the person smokes nowadays.

Value = 0 Label = No
Value = 1 Label = Yes

Main outcome in our analysis

"**kcigreg**" (ordinal): Frequency and amount smoked (8-15s) SC 8-15

Value = -9 Label = No answer/refused
Value = -8 Label = Don't know
Value = -6 Label = Schedule not obtained
Value = -2 Label = Schedule not applicable
Value = -1 Label = Item not applicable
Value = 1 Label = I have never smoked
Value = 2 Label = I have only smoked once or twice
Value = 3 Label = I used to smoke sometimes, but i never smoke a cigarette now
Value = 4 Label = I sometimes smoke, but i don't smoke every week
Value = 5 Label = I smoke between one and six cigarettes a week

Value = 6 Label = I smoke more than six cigarettes a week

Negative values were defined as Nas. Outcome for the additional analysis.

"**booklet**" (categorical): Which self-completion filled out.

Value = -9 Label = No answer/refused
Value = -8 Label = Don't know
Value = -6 Label = Schedule not obtained
Value = -2 Label = Schedule not applicable
Value = -1 Label = Item not applicable
Value = 1 Label = Pink 8-12
Value = 2 Label = Green 13-15
Value = 3 Label = Blue Young Adults
Value = 4 Label = White Adults

Negative values were defined as Nas. Filtered the sample according to adult self-completion booklet (4) or young self-completion booklet (1, 2). Blue young adults were excluded.

Other variables:

- "**weight**": scores adjusted due to the boost ethnicity sample (from *errorwt*).
- **"id":** year + pserial
- **"year":** 1999 (predictor)
- **"cigdyal":** frequency of smoking (adults). Used for additional analysis (smokers).

The description of the chosen variables as they were in the 2019 dataset are:

"**cignow_19**" (categorical): Whether smoke cigarettes nowadays (capi+casi)

Value = 1.0 Label = Yes
Value = 2.0 Label = No
Value = -9.0 Label = Refused
Value = -8.0 Label = Don't know
Value = -1.0 Label = Not applicable

Negative values were defined as Nas.

"**kcigregg**" (ordinal): Frequency of cigarette smoking (8-15s).

Value = -9 Label = No answer/refused
Value = -8 Label = Don't know
Value = -6 Label = Schedule not obtained
Value = -2 Label = Schedule not applicable
Value = -1 Label = Item not applicable
Value = 1 Label = Don't smoke cigarettes

Value = 2 Label = Smoke cigarettes, less than once a week
Value = 3 Label = Smoke cigarettes, once a week or more often

Negative values were defined as Nas. Values 2 and 3 were coded as smokers (1), while value 1 was coded for no-smokers (0).

"**smoke**" (categorical): Derived variable from cignow_19 (adult) and kcigregg (children) for 2019 dataset. Whether the person smokes nowadays.

Value = 0 Label = No
Value = 1 Label = Yes

Main outcome in our analysis.

"**KCigReg**" (ordinal): Frequency and amount smoked (8-15s) (SC)

Value = 1.0 Label = I have never smoked
Value = 2.0 Label = I have only smoked once or twice
Value = 3.0 Label = I used to smoke sometimes, but I never smoke a cigarette now
Value = 4.0 Label = I sometimes smoke, but I don't smoke every week
Value = 5.0 Label = I smoke between one and six cigarettes a week
Value = 6.0 Label = I smoke more than six cigarettes a week
Value = -9.0 Label = Refused
Value = -8.0 Label = Don't know
Value = -1.0 Label = Not applicable

Negative values were defined as Nas. Outcome for the additional analysis.

"**SCType**" (categorical): Type of self-completion offered.

Value = 1.0 Label = Age 16+
Value = 2.0 Label = Age 13-15
Value = 3.0 Label = Age 8-12
Value = -9.0 Label = Refused
Value = -8.0 Label = Don't know
Value = -1.0 Label = Not applicable

Negative values were defined as Nas. Filtered the sample according to young self-completion booklets (2, 3), or adult self-completion booklet (1).

Other variables:

- "**weight**": scores adjusted (from *wt_int*).
- **"id":** year + SerialA
- **"year":** 2019 (predictor)
- **"cigdyal":** frequency of smoking (adults). Used for additional analysis (smokers).